

Corrected Green MGGP Experiments: equal calls

Notas

- **O que mudou no MGGP:** o Green usa seleção de Pareto (NSGA-II), considerando erro e custo juntos. O modo MO-lex prioriza o erro e usa o custo para desempatar.
- **Correção comum:** no estudo novo, um avaliador externo corrige os atrasos de entrada e saída para todos os métodos. Mantém mínimos quadrados e os operadores; simulações que divergem são falhas, não zeros. O pacote original não foi editado.
- **Complexidade:** contamos as operações do polinômio sem termos repetidos: $C_\rho = a + \rho b$, com a adições e b multiplicações, incluindo os coeficientes. Isto é custo aritmético, não energia medida.
- **Dois pesos:** $\rho = 1$ conta todas as operações igualmente; $\rho_K = 729/64$ vem do modelo teórico em bits de Karatsuba. Os dois foram fixados antes das novas buscas e continuam na análise.
- **Medições:** tempos de execução permitem estimar $t_\times/t_+$; powermetrics mede potência para calcular energia; CodeCarbon fornece apenas a intensidade de carbono para calcular CO₂e. Energia e CodeCarbon não definem ρ .
- **Comparações novas:** além do protocolo original, todos os métodos receberam 4.510 avaliações por corrida. Também comparamos objetivos de número de termos e de nós, sem mudar a regra de escolha do modelo.
- **Estado deste relatório:** identificação corrigida concluída e energia dos novos modelos medida em uma sessão. As medições antigas não entram neste relatório.

1 Corrected evaluator and experimental protocol

This export uses corrected delayed-factor evaluation for every method. Least squares retains one column per gene, including duplicates; simulation uses the actual maximum factor delay for initialisation and complete windows of that delay plus $K = 20$. Divergence is a failure. The vendored package is unchanged; a shared external evaluator implements these corrections.

The comparison uses exactly 4,510 fitness calls per run: 100 initial evaluations and 49 steps with 90 evaluated offspring, including unchanged offspring and failures. Parent and survivor selection differ between methods; other representation and variation settings are shared. The mutation schedule uses the package’s nine-generation checkpoints.

NSGA-II supplies Pareto ranking and crowding [1]; the underlying multigene package is attributed to its software source [2]. Thirty seeds were run for each example and configuration under both weights. E1 is generated by

$$y(k) = 0.65y(k-1) + 0.35u(k) + 0.10u(k)y(k-1), \quad (1)$$

with white Gaussian measurement noise added in E2; E3 uses the Bouc–Wen simulator. The reference point remains (1, 1) and the declared cost reference is unchanged. Reference sensitivity

Table 1: Examples and search settings; C_{ref} for $\rho = 1$ (common: population 100, 50 generations, crossover 0.9, initial mutation 0.2; baseline elite retention 10%, multiple-shooting horizon $K = 20$, primitives `mul` and q^{-j}).

Example	N_{id}	N_{val}	genes	height	j_{max}	C_{ref}
E1 Tutorial (noise-free)	210	90	5	2	2	25
E2 Tutorial (noisy)	210	90	5	2	2	25
E3 Bouc–Wen hysteresis	300	200	8	3	3	72

is exported separately [3]. Holm correction includes all tested metrics, examples and baselines within each arm and weight (30 tests for the native arm, 60 for the equal-call arm). Run times under concurrent workloads, if present, are diagnostic rather than execution benchmarks.

2 Arithmetic complexity and declared weights

For each duplicate-free polynomial, $C_\rho = a + \rho b$: a counts additions and b counts multiplications, including coefficients. This theoretical operation model is evaluated directly from the structure. Both fixed scenarios are analysed: $\rho = 1$ counts operations equally; $\rho_K = 729/64$ is the Green-Box bit-level weight associated with Karatsuba [4, 5]. Neither comes from measured power or CodeCarbon. The structural invariance and cost properties remain analytical results; population hashes and conditional hypervolume checks test the corresponding search statements empirically.

Timing calibration fits $t = t_0 + t_+a + t_\times b$ and estimates $t_\times/t_+$ for the chosen execution implementation. Earlier Python timing measurements motivate an order-one primary weight; both weights were fixed before these corrected searches. The separate C microbenchmark is a diagnostic, not a calibration of CPython and not a reason to retune the objectives retrospectively. Powermetrics supplies power for energy integration; CodeCarbon supplies only grid carbon intensity. Corrected energy measurements are reported in the final section.

3 Results under both weights

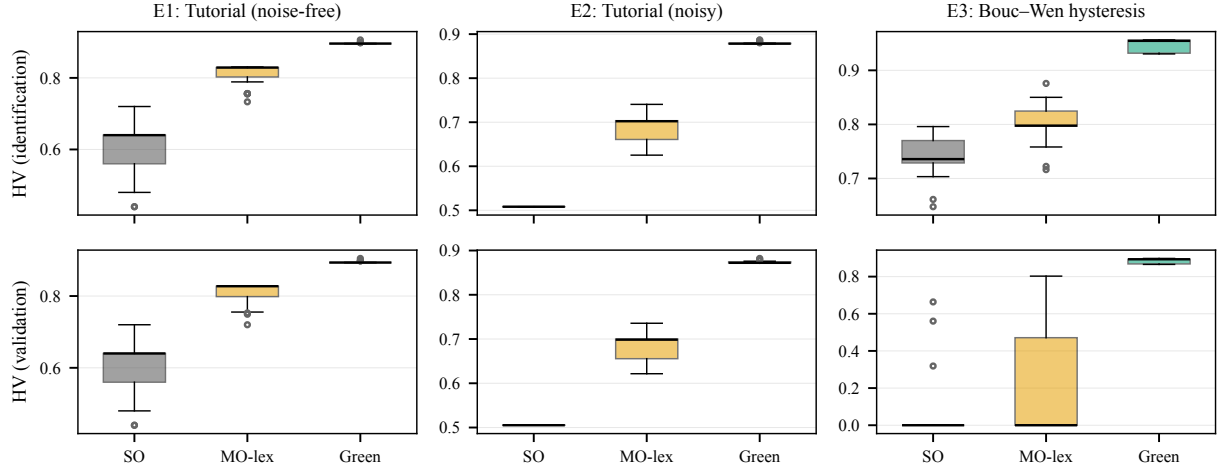


Figure 1: Corrected results: hv.

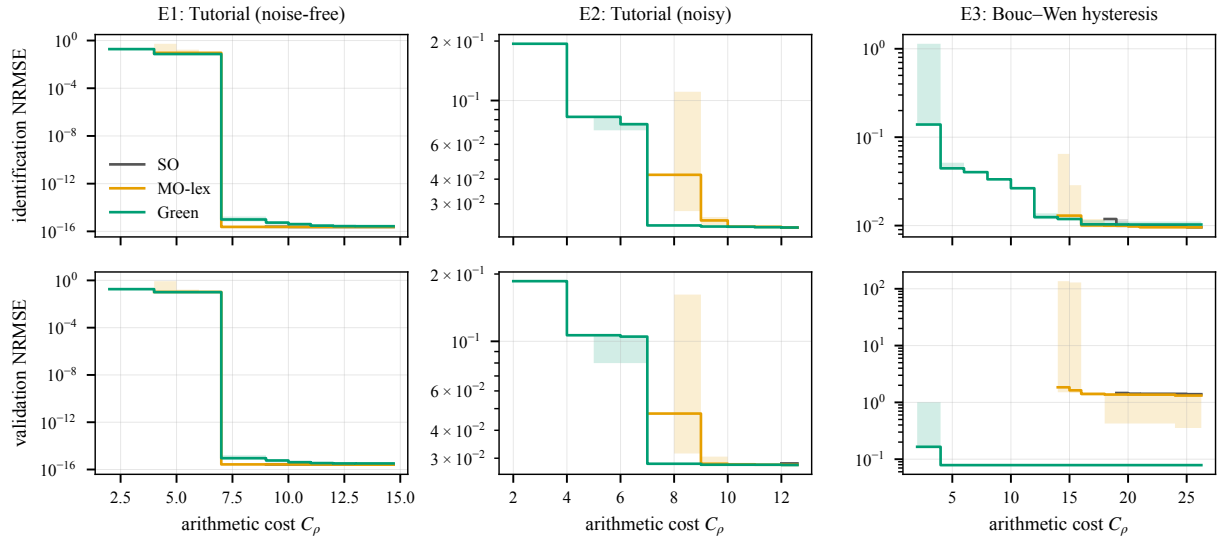


Figure 2: Corrected results: attainment.

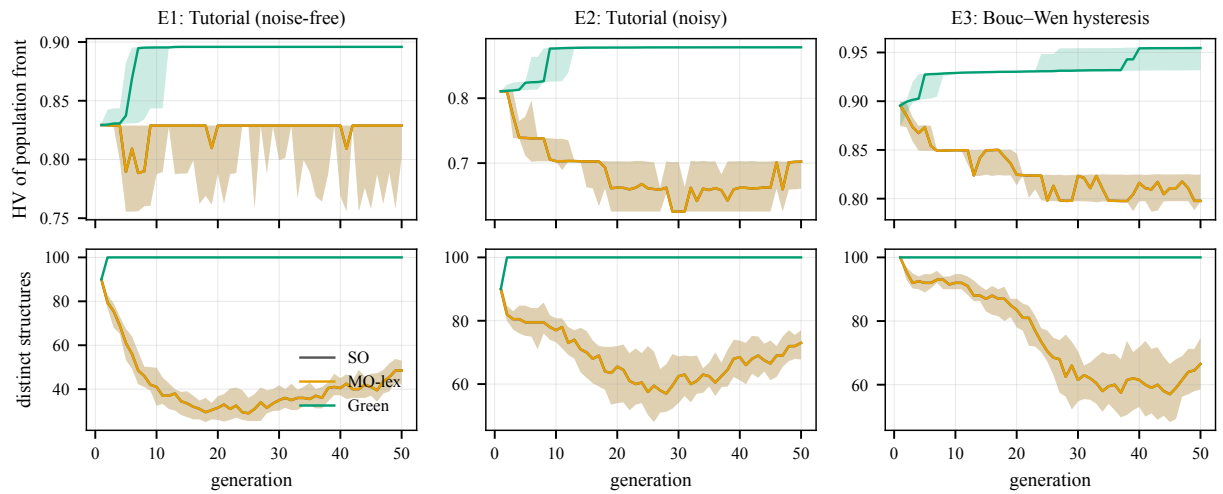


Figure 3: Corrected results: convergence.

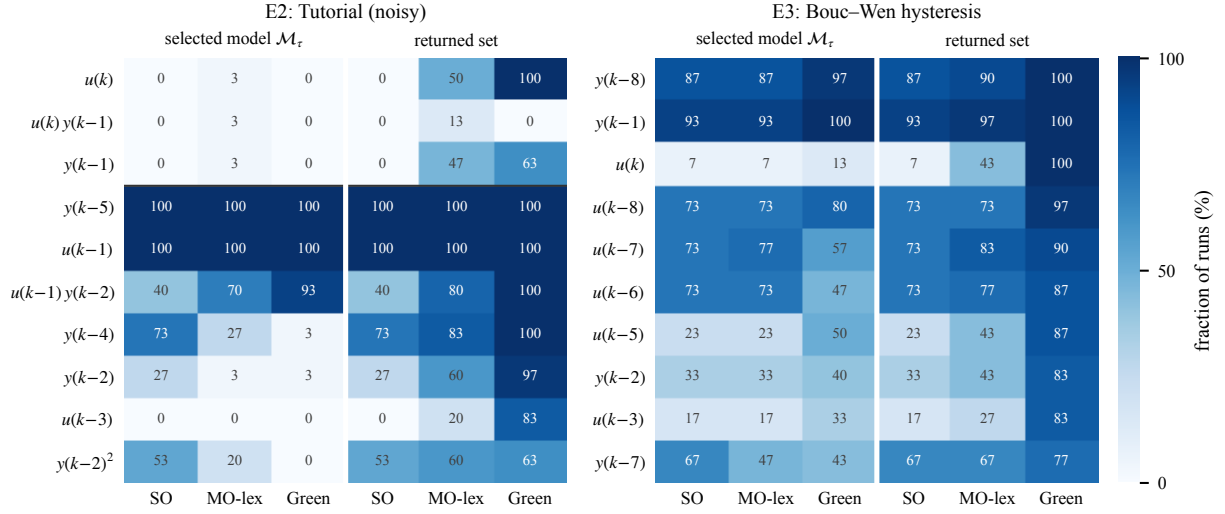


Figure 4: Corrected results: regressors.

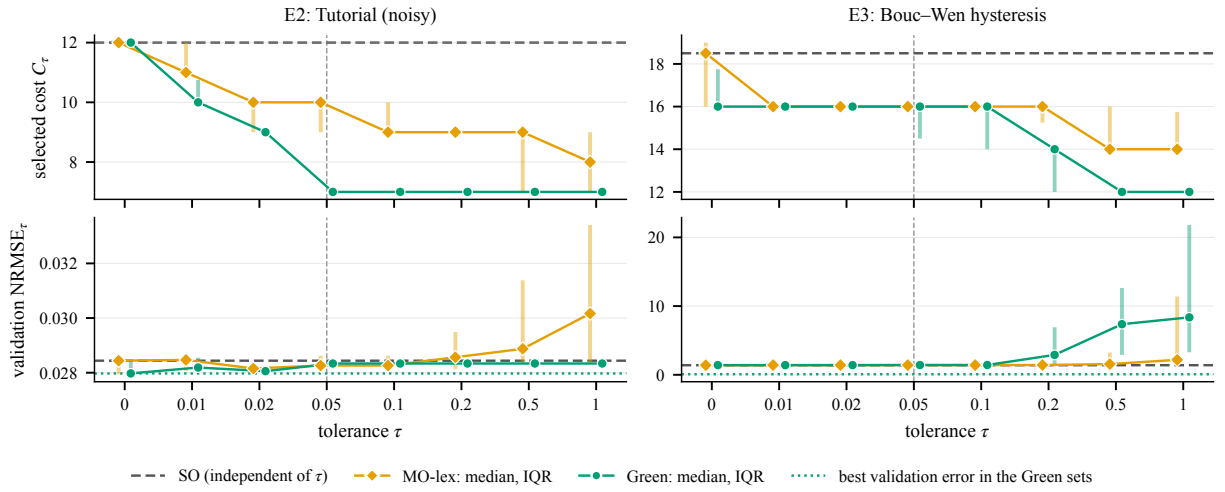


Figure 5: Corrected results: tau.

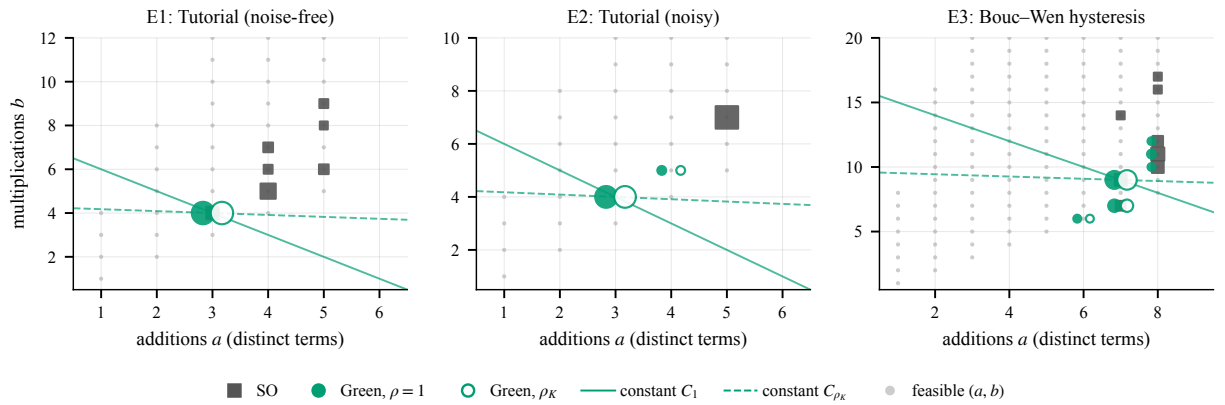


Figure 6: Corrected results: ab.

Table 2: Results over 30 independent runs with $\rho = 1$: median [interquartile range]. HV: hypervolume of the delivered set (identification or free-run validation error); C_τ , NRMSE_τ : cost and free-run validation NRMSE of the model selected by the τ -rule; Rec_τ , $\text{Rec}_\mathcal{P}$: fraction of runs in which the selected model, or at least one model of the returned set, contains the three true regressors of (1). $^*/^\dagger$: Green differs from SO/MO-lex (Mann–Whitney, Holm-adjusted $p < 0.05$).

Ex.	Method	HV _{id}	HV _{val}	C_τ	NRMSE_τ	Rec_τ	$\text{Rec}_\mathcal{P}$
E1	SO	0.640 [0.560, 0.640]	0.640 [0.560, 0.640]	9 [9, 11]	3e-16 [2e-16, 3e-16]	1.00	1.00
	MO-lex	0.829 [0.803, 0.829]	0.827 [0.798, 0.827]	7 [7, 7]	3e-16 [3e-16, 3e-16]	1.00	1.00
	Green	0.896 [0.896, 0.896] *†	0.893 [0.893, 0.893] *†	7 [7, 7] *	9e-16 [7e-16, 2e-15] *†	1.00	1.00
E2	SO	0.508 [0.508, 0.508]	0.505 [0.505, 0.505]	12 [12, 12]	0.028 [0.028, 0.028]	0.00	0.00
	MO-lex	0.702 [0.661, 0.703]	0.699 [0.656, 0.700]	10 [9, 10]	0.028 [0.028, 0.029]	0.00	0.07
	Green	0.878 [0.878, 0.879] *†	0.872 [0.872, 0.874] *†	7 [7, 7] *†	0.028 [0.028, 0.028]	0.00	0.00
E3	SO	0.736 [0.729, 0.770]	0.000 [0.000, 0.000]	18.5 [16, 19]	1.394 [1.301, 1.463]	–	–
	MO-lex	0.798 [0.797, 0.825]	0.000 [0.000, 0.471]	16 [16, 16]	1.411 [1.340, 1.467]	–	–
	Green	0.955 [0.932, 0.956] *†	0.893 [0.868, 0.893] *†	16 [14.5, 16] *	1.420 [1.316, 1.451]	–	–

Table 3: Regressors of the model selected by the τ -rule in the representative run (median validation NRMSE of the selected model) and its free-run validation NRMSE.

Ex.	Method	Regressors	C	NRMSE
E2	SO	$u(k)y(k-2)$, $u(k-1)$, $y(k-2)$, $y(k-2)^2$, $y(k-5)$	12	0.028
E2	MO-lex	$u(k-1)$, $u(k-1)y(k-2)$, $y(k-5)$	7	0.028
E2	Green	$u(k-1)$, $u(k-1)y(k-2)$, $y(k-5)$	7	0.028
E3	SO	$u(k-2)u(k-4)y(k-3)$, $u(k-6)$, $u(k-7)$, $u(k-8)$, $y(k-1)$, $y(k-2)$, $y(k-7)$, $y(k-8)$	18	1.413
E3	MO-lex	$u(k-3)u(k-4)y(k-2)$, $u(k-6)$, $u(k-7)$, $u(k-8)$, $y(k-1)$, $y(k-2)$, $y(k-8)$	16	1.418
E3	Green	$u(k)$, $u(k-1)$, $u(k-2)$, $u(k-3)$, $y(k-1)$, $y(k-7)$, $y(k-8)$	14	1.425

Table 4: Empirical verification of Theorems 2 and 3 over 30 seeds. Lexicographic reduction: seeds in which SO and MO-lex produce identical populations in all generations; diverging seeds in which the divergence is preceded by an error tie between individuals of different cost, and in which the populations involved contain at least two individuals with $J_E = \infty$; median generation of the first divergence. Non-deterioration: largest $|U_t^1|$ over all runs and generations, bound $L(G, h)$ and population size μ ; runs with at least one decrease of the population-front hypervolume.

Ex.	Lexicographic reduction				Capacity			Runs with HV decrease		
	identical	tie	$J_E = \infty$	first div.	$\max U_t^1 $	$L(G, h)$	μ	SO	MO-lex	Green
E1	30/30	0/0	0/0	–	8	50	100	30	30	0
E2	30/30	0/0	0/0	–	9	50	100	30	30	0
E3	30/30	0/0	0/0	–	10	260	100	30	30	0

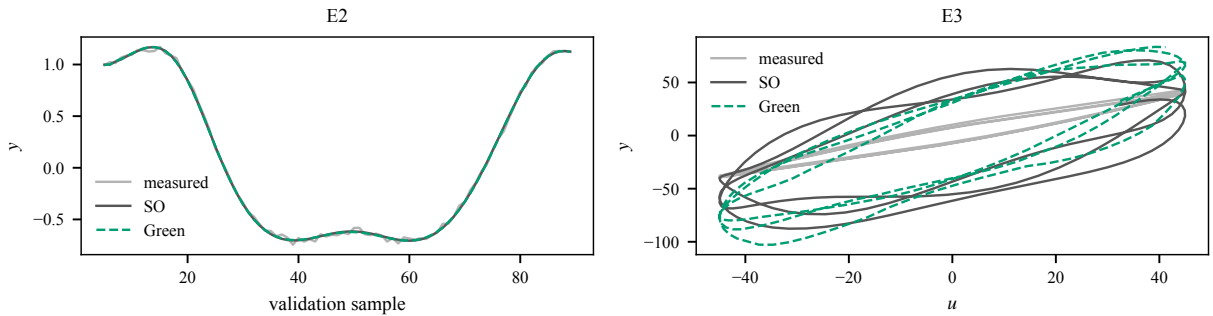


Figure 7: Corrected results: freerun.

Table 5: Effect of the multiplication weight ρ over 30 runs (medians). HV: identification hyper-volume; ΔC_τ : reduction of the median selected cost of Green MGGP relative to SO, with the cost of the same scenario; $a+b$ and C_{ρ_K} : operation count and Karatsuba-weighted cost of the selected Green model; same: fraction of seeds in which Green MGGP selects the same structure under both weights.

Ex.	weight	HV _{id}			ΔC_τ	Green $\mathcal{M}_\tau$		
		SO	MO-lex	Green		$a+b$	C_{ρ_K}	same
E1	$\rho = 1$	0.640	0.829	0.896	22%	7	48.6	100%
	ρ_K	0.738	0.884	0.929	20%	7	48.6	
E2	$\rho = 1$	0.508	0.702	0.878	42%	7	48.6	80%
	ρ_K	0.622	0.773	0.910	43%	7	48.6	
E3	$\rho = 1$	0.736	0.798	0.955	14%	16	109.5	0%
	ρ_K	0.819	0.874	0.969	14%	16	109.5	

4 Execution Time, Energy and Carbon of the Corrected Models

This separate stage reads the equal-evaluation caches and measures execution of their selected models. It does not alter C_ρ , either declared weight or the τ -rule. Historical energy measurements are not included.

4.1 Reconstruction and execution implementations

The 450 selected records comprise SO, MO-lex and Green under $\rho = 1$, and MO-lex and Green under ρ_K , across three examples and 30 seeds. Four failed validations remain missing; 446 models were rebuilt, re-estimated and checked against their saved costs and validation errors. The primary canonical simulator evaluates the duplicate-free polynomial in straight-line Python with exactly a additions and b multiplications per sample. The secondary NumPy simulator evaluates every gene with corrected delays; it is not the unmodified package predictor. The two implementations agree within the declared reconstruction tolerances.

4.2 Timing, power integration and carbon conversion

Each model’s timing is the median of five repeated-simulation batches lasting at least 50 ms. Power was measured in a separate mains-powered session on an Apple M2 with Python 3.13.9, using CPU, GPU and neural-engine power from `powermetrics` at 100 ms intervals. Ten rounds each visit 30 groups in a fixed seeded random order, with complete cycles over the valid models of each group for at least 10 s. Eleven idle blocks bracket the rounds, giving 311 blocks in total. Every block has valid power samples.

Net energy per simulation is

$$E_{\text{sim}} = \frac{E_{\text{block}} - \tilde{P}_{\text{idle}} t_{\text{block}}}{n_{\text{sim}}}, \quad (2)$$

where the baseline is the median power across all idle blocks. Idle block means range from 0.056 to 0.745 W. Replacing the baseline by the mean of the two idle blocks bracketing each round leaves all Holm-significance decisions unchanged in the sensitivity analysis. CodeCarbon supplies only the Irish grid intensity, 290.805 gCO₂e/kWh [6]; emissions are $E_{\text{sim}} I_{\text{grid}}$ after conversion to kWh. CodeCarbon does not track energy inside the measurement blocks and neither power nor carbon intensity establishes ρ .

Energy ratios to SO are formed within each round. Green is compared with SO, MO-lex and Green-K using two-sided Mann–Whitney tests over rounds, with Holm correction across the 18 example–simulator–comparator tests and A_{12} . These are within-session comparisons, not independent hardware sessions.

4.3 Time-based assessment of the fixed weights

For 127 distinct canonical structures, the fit $t = t_0 + t_+ a + t_- b$ gives $t_+ = 10.16$ ns, $t_- = 9.21$ ns and $R^2 = 0.9986$. The ratio $t_-/t_+ = 0.906$ is close to one for this Python implementation. The operation-count fit has $R^2 = 0.9984$, compared with 0.9488 for the Karatsuba-weighted cost. These measurements support the scale of the already fixed unit weight; they do not retune the completed searches. The coefficients include interpreter overhead, are fitted to search-selected structures and do not measure instruction-level costs. The separate C microbenchmark is a diagnostic of another execution implementation.

4.4 Energy results and their limits

Table 6 gives absolute time, energy and carbon estimates, while Fig. 9 compares energy relative to SO within each round. In that figure, values below one indicate savings and values above one an

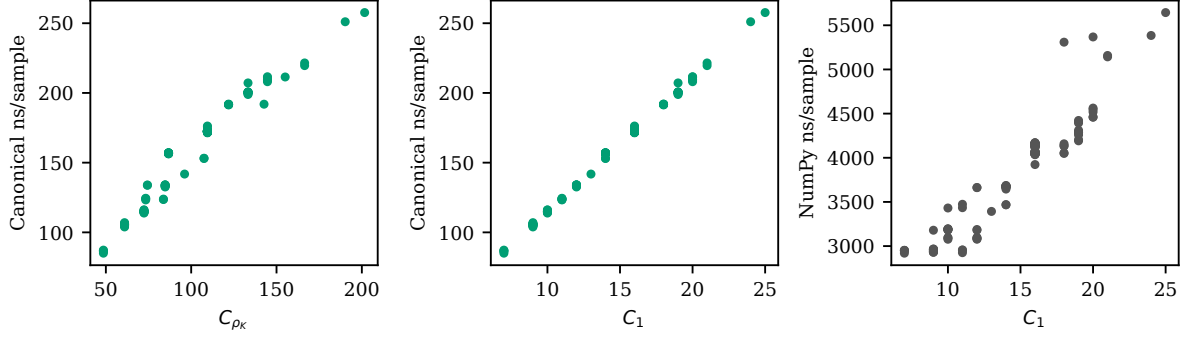


Figure 8: Corrected-model execution time per sample: canonical simulation against C_{ρ_K} and C_1 , and corrected all-gene NumPy simulation against C_1 . Both search weights are represented.

Table 6: Execution of the corrected selected models: median time, net energy and estimated carbon per simulation over ten measurement rounds. Canonical execution evaluates distinct monomials; all-gene NumPy execution uses corrected delays. Suffix K denotes searches under ρ_K .

Ex.	Method	Canonical polynomial			All-gene NumPy		
		Time (μ s)	Energy (μ J)	CO ₂ e (μ g)	Time (μ s)	Energy (μ J)	CO ₂ e (μ g)
E1	SO	9.91	65.4	0.00528	313	2.11e+03	0.17
E1	MO-lex	7.78	51.6	0.00417	329	2.31e+03	0.186
E1	Green	7.79	52.2	0.00422	337	2.37e+03	0.192
E1	MO-lex-K	7.82	52.5	0.00424	334	2.32e+03	0.187
E1	Green-K	7.78	51.1	0.00413	330	2.35e+03	0.19
E2	SO	11.7	76.7	0.0062	338	2.34e+03	0.189
E2	MO-lex	9.49	65.7	0.0053	328	2.24e+03	0.181
E2	Green	7.66	51.2	0.00414	324	2.23e+03	0.18
E2	MO-lex-K	9.44	62.7	0.00507	326	2.23e+03	0.18
E2	Green-K	7.65	50.4	0.00407	321	2.25e+03	0.182
E3	SO	38.4	251	0.0203	848	5.8e+03	0.469
E3	MO-lex	35.3	232	0.0188	832	5.55e+03	0.448
E3	Green	33.6	228	0.0184	789	5.26e+03	0.425
E3	MO-lex-K	35.4	233	0.0188	830	5.6e+03	0.452
E3	Green-K	32.9	219	0.0177	783	5.38e+03	0.434

increase; the bars describe variation over rounds in this session. For canonical execution, median per-round Green/SO energy ratios are 0.798, 0.665 and 0.893 on E1–E3. The corresponding reductions are 20.2%, 33.5% and 10.7%. Reductions are significant against SO on E1 and E2 (Holm-adjusted $p = 0.003$ and $p = 0.003$), whereas the E3 difference is not significant ($p = 0.542$). Green differs significantly from MO-lex in canonical energy only on E2, and not from Green-K on any example after Holm correction.

The secondary implementation changes the outcome: on E1 Green uses 11.8% more energy than SO ($p = 0.031$). E2 shows no significant difference against SO ($p = 1.000$); on E3 Green uses 7.9% less energy ($p = 0.009$), and also differs from MO-lex. Arithmetic savings of the canonical polynomial therefore do not guarantee savings when all genes are executed through NumPy. These findings concern interpreted execution on one laptop system-on-chip in one

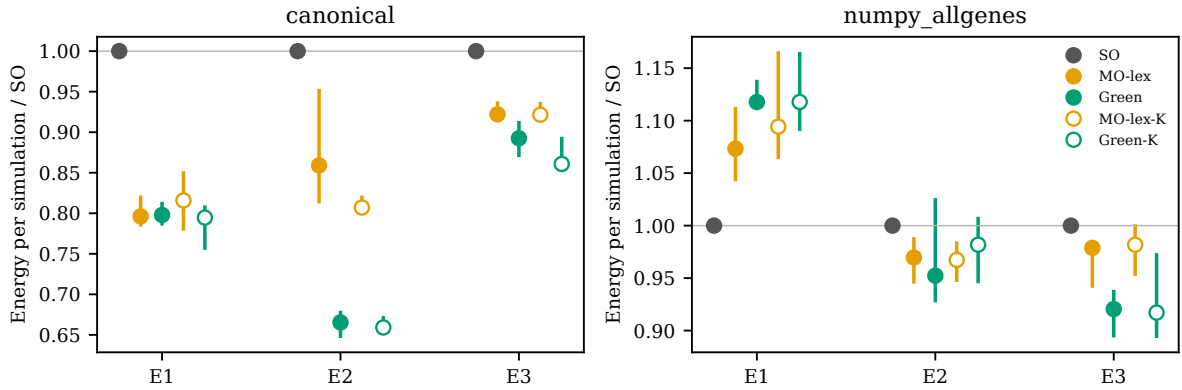


Figure 9: Corrected selected-model energy relative to SO within each of ten rounds: medians and interquartile ranges. Open markers denote the Karatsuba-weighted searches; the secondary simulator uses corrected delays.

session. They do not establish embedded-hardware performance or a universal multiplication weight. E3 selected models generalise poorly under the unchanged identification-only rule, so energy reductions cannot be presented as savings at preserved validation accuracy. Carbon values are energy-derived estimates using one grid-intensity factor, not directly measured emissions.

Ablation and interpretation

The term-count ablation has no significant difference in validation hypervolume from Green on any tested example after Holm correction. These results do not establish a consistent advantage of the arithmetic objective over term count. On E1, validation errors are at machine precision, so statistical distinctions below 10^{-14} have no practical accuracy interpretation. On E3, the identification-only τ -rule selects models with poor free-run generalisation even though the Green sets contain models with substantially better validation. The rule remains unchanged; the best available validation is a diagnostic and is not used for selection.

Table 7: Corrected equal-evaluation ablation: medians over 30 runs with $\rho = 1$.

example	config	HV (id)	HV (val)	Cost	NRMSE	Evaluations
E1	Green	0.896	0.893	7.000	0.000	4510.000
	MO-lex	0.829	0.827	7.000	0.000	4510.000
	Pareto-nodes	0.831	0.828	7.000	0.000	4510.000
	Pareto-terms	0.896	0.893	7.000	0.000	4510.000
	SO	0.640	0.640	9.000	0.000	4510.000
E2	Green	0.878	0.872	7.000	0.028	4510.000
	MO-lex	0.702	0.699	10.000	0.028	4510.000
	Pareto-nodes	0.812	0.807	11.000	0.028	4510.000
	Pareto-terms	0.878	0.872	7.000	0.028	4510.000
	SO	0.508	0.505	12.000	0.028	4510.000
E3	Green	0.955	0.893	16.000	1.420	4510.000
	MO-lex	0.798	0.000	16.000	1.411	4510.000
	Pareto-nodes	0.929	0.868	14.000	1.425	4510.000
	Pareto-terms	0.954	0.893	16.000	1.189	4510.000
	SO	0.736	0.000	18.500	1.394	4510.000

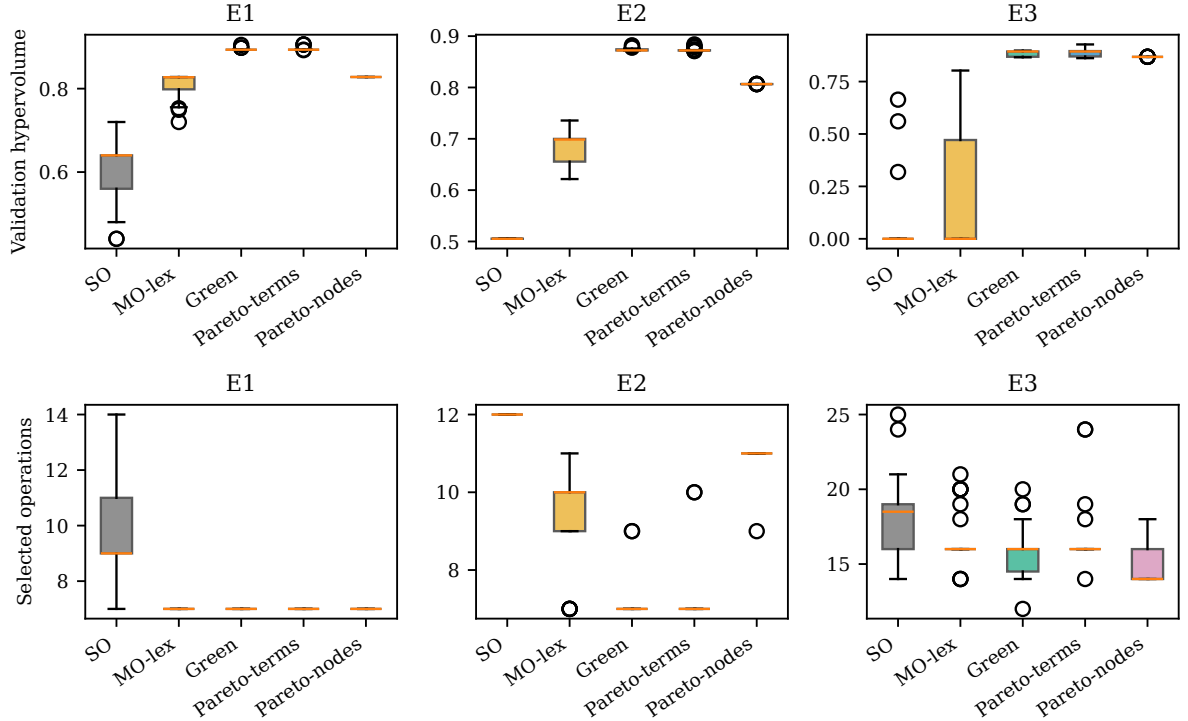


Figure 10: Equal-evaluation comparison with term and node objectives: validation hypervolume and selected arithmetic cost over 30 runs.

References

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